

Deep Decision Optimization

A regularized-duality view of SPO+

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Scientific question

Can we preserve the SPO+ regret-surrogate structure while replacing its hard support-function subgradient by a smooth, compressed moment map?

Core Improvement Highlight

Duality language

SPO+ uses a **hard support-function subgradient**

= an exposed-face selector of the feasible region.

DDO replaces it with a **regularized support moment**

= the gradient of a regularized Fenchel conjugate.

Roadmap

1. Predict-then-optimize and SPO+
2. SPO+ as a support-function / Fenchel-duality object
3. DDO as regularized support-function smoothing
4. Why exact smoothing can be exponential
5. Plackett–Luce primer: the clean compression example
6. General DDO abstraction: compressed moment oracle
7. Proof obligations and experiments

Predict-then-optimize setup

For context x , predict a cost vector

$$\hat{c} = f_{\theta}(x).$$

Then solve a downstream optimization problem

$$w^*(\hat{c}) \in \arg \min_{w \in S} \hat{c}^{\top} w.$$

The realized cost vector is c .

The decision regret / SPO loss is

$$\ell_{\text{SPO}}(\hat{c}, c) = c^{\top} w^*(\hat{c}) - c^{\top} w^*(c).$$

Why raw SPO is hard

The map

$$\hat{c} \mapsto w^*(\hat{c})$$

is often piecewise constant.

Therefore raw SPO loss is usually:

- flat inside decision regions;
- discontinuous or nonsmooth at boundaries;
- difficult for gradient-based deep learning.

Motivation

This is exactly why SPO+ was introduced.

SPO+ formula

SPO+ is

$$\ell_{\text{SPO}+}(\hat{c}, c) = \max_{w \in S} \{(c - 2\hat{c})^\top w\} + 2\hat{c}^\top w^*(c) - c^\top w^*(c).$$

Let

$$u = c - 2\hat{c}.$$

The key nonsmooth term is

$$\max_{w \in S} u^\top w.$$

Goal

Understand and smooth this term without changing the SPO+ backbone.

Move to convex-duality notation

Let

$$C = \text{conv}(S).$$

For a linear objective, optimizing over S and C gives the same value.

Define the indicator function

$$\delta_C(w) = \begin{cases} 0, & w \in C, \\ +\infty, & w \notin C. \end{cases}$$

The support function is

$$\sigma_C(u) = \max_{w \in C} u^\top w.$$

Support function = Fenchel conjugate

The support function is the Fenchel conjugate of the feasible-set indicator:

$$\sigma_C(u) = \delta_C^*(u) = \max_{w \in C} \{u^\top w - \delta_C(w)\}.$$

Therefore SPO+ can be written as

$$\ell_{\text{SPO}+}(\hat{c}, c) = \sigma_C(c - 2\hat{c}) + 2\hat{c}^\top w^*(c) - c^\top w^*(c).$$

The dual direction is

$$u = c - 2\hat{c}.$$

The hard support selector

The support-function subdifferential is

$$\partial\sigma_C(u) = \arg \max_{w \in C} u^\top w.$$

This set is the **exposed face** of C in direction u :

$$F_C(u) = \{w \in C : u^\top w = \sigma_C(u)\}.$$

So SPO+ uses an exposed-face selector:

$$w_{\text{sup}}(u) \in \partial\sigma_C(u).$$

SPO+ subgradient

By the chain rule with $u = c - 2\hat{c}$,

$$\partial_{\hat{c}} \ell_{\text{SPO}+}(\hat{c}, c) = 2w^*(c) - 2\partial\sigma_C(c - 2\hat{c}).$$

If the support maximizer is unique,

$$\nabla_{\hat{c}} \ell_{\text{SPO}+} = 2[w^*(c) - w_{\text{sup}}(c - 2\hat{c})].$$

Using the minimization oracle,

$$w_{\text{sup}}(c - 2\hat{c}) = w^*(2\hat{c} - c).$$

Geometric interpretation

For the hard support selector,

$$w \in \partial\sigma_C(u) \iff u \in N_C(w),$$

where $N_C(w)$ is the normal cone of C at w .

So SPO+ selects a point or face whose normal cone contains the dual direction u .

Remaining algorithmic sharpness

For polyhedral C , this selector jumps between vertices/faces as u moves.

Clean DDO question

SPO+ is already a principled decision-regret surrogate.

Narrow DDO question

Can we replace

$$\sigma_C(u) = \delta_C^*(u)$$

by a smooth regularized conjugate

$$(\delta_C + \tau\Omega)^*(u)$$

while recovering SPO+ as $\tau \downarrow 0$?

Regularized support function

Choose a mirror potential / regularizer Ω on C .

Define

$$\sigma_{C,\tau}^{\Omega}(u) = (\delta_C + \tau\Omega)^*(u) = \max_{w \in C} \{u^{\top} w - \tau\Omega(w)\}.$$

Then define the DDO-SPO+ loss

$$\ell_{\text{DDO}}^{\tau,\Omega}(\hat{c}, c) = \sigma_{C,\tau}^{\Omega}(c - 2\hat{c}) + 2\hat{c}^{\top} w^*(c) - c^{\top} w^*(c).$$

Regularized selector / moment map

If Ω is strongly convex, the maximizer is unique:

$$w_\tau(u) = \nabla \sigma_{C,\tau}^\Omega(u) = \arg \max_{w \in C} \{u^\top w - \tau \Omega(w)\}.$$

The DDO gradient is

$$\nabla_{\hat{c}} \ell_{\text{DDO}}^{\tau,\Omega} = 2 [w^*(c) - w_\tau(c - 2\hat{c})].$$

Interpretation

SPO+ uses a hard exposed-face selector. DDO uses a regularized support selector.

Normal-cone smoothing

Hard SPO+ support selection satisfies

$$u \in N_C(w).$$

Regularized DDO support selection satisfies the KKT condition

$$u - \tau \nabla \Omega(w_\tau) \in N_C(w_\tau).$$

Dual geometry is preserved

DDO does not discard the dual geometry. It smooths the normal-cone selection rule by adding a mirror correction term.

Zero-temperature recovery

If the regularization term vanishes,

$$\sigma_{C,\tau}^{\Omega}(u) \rightarrow \sigma_C(u) \quad \text{as } \tau \downarrow 0.$$

Therefore

$$\ell_{\text{DDO}}^{\tau,\Omega}(\hat{c}, c) \rightarrow \ell_{\text{SPO}+}(\hat{c}, c).$$

Conservative property

SPO+ is the zero-temperature endpoint of DDO.

Finite-decision entropy smoothing

Suppose

$$S = \{w_1, \dots, w_K\}.$$

The entropic smoothed support is

$$\phi_{\tau,S}(u) = \tau \log \sum_{k=1}^K \exp(u^\top w_k / \tau).$$

It has the variational form

$$\phi_{\tau,S}(u) = \max_{q \in \Delta_K} \left\{ \sum_{k=1}^K q_k u^\top w_k - \tau \sum_{k=1}^K q_k \log q_k \right\}.$$

Entropic support moment

The optimizer over q is

$$q_k(u) = \frac{\exp(u^\top w_k / \tau)}{\sum_{j=1}^K \exp(u^\top w_j / \tau)}.$$

The gradient is the soft support moment

$$\nabla_u \phi_{\tau, S}(u) = \sum_{k=1}^K q_k(u) w_k.$$

Soft exposed-face replacement

DDO replaces an exposed face by a probability-weighted moment over nearby exposed decisions.

Approximation guarantee

For finite S ,

$$\sigma_C(u) \leq \phi_{\tau,S}(u) \leq \sigma_C(u) + \tau \log |S|.$$

Thus

$$\ell_{\text{SPO}+} \leq \ell_{\text{DDO}}^\tau \leq \ell_{\text{SPO}+} + \tau \log |S|.$$

Controlled smoothing bias

The bias disappears as $\tau \downarrow 0$.

Curvature information

For entropic smoothing,

$$\nabla_u^2 \phi_{\tau,S}(u) = \frac{1}{\tau} \text{Cov}_{q(u)}(w).$$

This tells us which feasible decisions are near-competitive in the support direction.

Hard SPO+

One exposed-face selector.

DDO

mean $\nabla \phi_{\tau,S}(u)$, covariance $\nabla^2 \phi_{\tau,S}(u)$.

Why exact smoothing is not enough

The formula

$$\phi_{\tau,S}(u) = \tau \log \sum_{w \in S} \exp(u^\top w / \tau)$$

requires summing over all feasible decisions.

For many downstream problems, $|S|$ is exponential.

For rankings,

$$|S| = N!.$$

Second idea needed

compressed computation of the support moment.

Why introduce Plackett–Luce?

Plackett–Luce is **not** the universal DFL assumption.
It is the cleanest example of the compression principle.

Compression principle

Do not maintain a distribution over all $N!$ rankings. Maintain N scores whose mirror map induces a ranking distribution.

This is the useful RIPLM idea for DDO.

PL primer: the object being compressed

A ranking is a permutation

$$\sigma = (\sigma_1, \dots, \sigma_N).$$

A fully general distribution over rankings needs one probability for each permutation:

$$q(\sigma), \quad \sigma \in \mathfrak{S}_N, \quad |\mathfrak{S}_N| = N!.$$

This is the exponential object

A naive distributional representation has $O(N!)$ degrees of freedom.

PL primer: score parameterization

PL uses one score per item:

$$s = (s_1, \dots, s_N) \in \mathbb{R}^N.$$

At position t , choose from remaining items R_t :

$$\mathbb{P}(\sigma_t = i \mid R_t) = \frac{\exp(s_i/\tau)}{\sum_{j \in R_t} \exp(s_j/\tau)}.$$

The full ranking probability is

$$\mathbb{P}_s(\sigma) = \prod_{t=1}^N \frac{\exp(s_{\sigma_t}/\tau)}{\sum_{j \in R_t} \exp(s_j/\tau)}.$$

What PL compresses

The distribution still lives on $N!$ rankings.

But it is represented by

N scores

rather than

$N!$ free probabilities.

Ranking version of DDO principle

large decision distribution $\rightarrow$ low-dimensional dual coordinates.

One-step PL geometry = softmax geometry

For a one-step choice from an available set E_t ,

$$p_t(i) = \frac{\exp(s_i/\tau_t) \mathbf{1}\{i \in E_t\}}{\sum_{j \in E_t} \exp(s_j/\tau_t)}.$$

This is the gradient of the log-partition function

$$\Psi_{\tau_t}(s) = \tau_t \log \sum_{j \in E_t} \exp(s_j/\tau_t) :$$

$$p_t = \nabla \Psi_{\tau_t}(s).$$

Entropy duality behind softmax

The conjugate of the log-partition map is negative entropy on the simplex:

$$\Psi_{\tau}^*(p) = \tau \sum_i p_i \log p_i + \delta_{\Delta}(p).$$

Thus score-space updates correspond to mirror descent with KL / entropy geometry.

Key point

PL is not merely a probability model. It is also a tractable mirror map from scores to decision probabilities.

Centered residual identity

Let

$$L(s) = \sum_i p_i(s) \ell_i, \quad p = \nabla \Psi_\tau(s).$$

Then

$$\frac{\partial L}{\partial s_i} = \frac{p_i}{\tau} (\ell_i - \bar{\ell}), \quad \bar{\ell} = \sum_j p_j \ell_j.$$

The gradient is a centered residual:

- better-than-average items are pushed up;
- worse-than-average items are pushed down.

RIPLM algorithmic lesson

The relevant lesson from RIPLM is not

all DFL problems are rankings.

Useful lesson

Represent the soft decision distribution in mirror-dual coordinates, then update those coordinates using residual information.

For ranking, this replaces an $O(N!)$ distributional representation by an $O(N)$ score representation.

What exactly is $O(N!) \rightarrow O(N)$?

Naive ranking distribution:

$$q(\sigma) \quad \text{for each } \sigma \in \mathfrak{S}_N \quad \Rightarrow \quad O(N!) \text{ parameters.}$$

PL/RIPLM score representation:

$$s_1, \dots, s_N \quad \Rightarrow \quad O(N) \text{ parameters.}$$

Nuance

Full ranking generation may still involve sequential choices, but the maintained geometry is N -dimensional.

From PL to general DDO

For general DFL, we cannot assume PL.

The abstraction is a structured family of decision distributions

$$\mathcal{Q} = \{q_s(w) : s \in \mathbb{R}^d\}.$$

The object DDO needs is not the full distribution.

It needs the first moment

$$M_\tau(u) = \mathbb{E}_{w \sim q_s}[w].$$

This is the **compressed support moment oracle**.

General DDO support-moment oracle

Replace

$$\partial\sigma_C(u)$$

by

$$M_\tau(u) \approx \nabla\sigma_{C,\tau}^\Omega(u).$$

Then the training gradient is

$$\nabla_{\hat{c}}\ell_{\text{DDO}} = 2[w^*(c) - M_\tau(c - 2\hat{c})].$$

The algorithmic cost becomes

$$O(|S|) \rightsquigarrow O(T_{\text{moment}}).$$

Examples of moment oracles

Downstream structure	Hard object	Possible soft moment oracle
Ranking	$N!$ rankings	PL / score-space softmax
Shortest path	exponential paths	soft dynamic programming / edge marginals
Assignment	$N!$ matchings	entropy-regularized assignment / Sinkhorn
Black-box solver	unknown $\mathcal{S}$	candidate pool + log-sum-exp

Science question

Does the moment oracle preserve enough of the support geometry to improve training?

Ranking DDO instance

If feasible decisions are rankings, use a PL-style moment oracle.

The support direction u induces item-level scores.

The PL mirror map gives probabilities without enumerating $N!$ rankings.

Then DDO uses

$$M_{\tau}(u) = \mathbb{E}_{\sigma \sim \text{PL}(s(u))}[w(\sigma)].$$

Key contribution

Compressed mirror computation of the support moment.

Shortest-path DDO instance

For paths,

$$\phi_\tau(u) = \tau \log \sum_{y \in \mathcal{P}} \exp(u^\top y / \tau).$$

Naively, $\mathcal{P}$ can be exponential.

On a DAG or grid, soft dynamic programming can compute:

$$\phi_\tau(u) \quad \text{and} \quad M_\tau(u) = \nabla \phi_\tau(u).$$

The moment $M_\tau(u)$ is an edge-marginal vector.

Candidate-set DDO instance

If there is no clean structure, use a local candidate set

$$\mathcal{C}_i \subset S.$$

For example:

- true optimal decision $w^*(c_i)$;
- SPO+ support selector $w^*(2\hat{c}_i - c_i)$;
- perturbed support selectors;
- top- k solver outputs;
- cached historical feasible decisions.

Then

$$\phi_{\tau, \mathcal{C}_i}(u) = \tau \log \sum_{w \in \mathcal{C}_i} \exp(u^\top w / \tau).$$

DDO-SPO+ training algorithm

For each example (x_i, c_i) :

1. Predict

$$\hat{c}_i = f_{\theta}(x_i).$$

2. Compute the true-cost optimizer

$$w_i^+ = w^*(c_i).$$

3. Form the SPO+ dual direction

$$u_i = c_i - 2\hat{c}_i.$$

4. Compute the regularized support moment

$$m_i = M_{\tau}(u_i).$$

5. Backpropagate

$$g_{\hat{c}_i} = 2(w_i^+ - m_i).$$

DDO-MD variant

There is also a more RIPLM-like variant.

Instead of smoothing the SPO+ support term, define a soft decision distribution

$$p_{\theta}(w) \propto \exp(-\hat{c}^{\top} w / \tau)$$

and train expected realized cost

$$\mathcal{R}_{\tau}(\hat{c}, c) = \mathbb{E}_{p_{\theta}}[c^{\top} w].$$

Its gradient has a centered residual form:

$$\nabla_{\hat{c}} \mathcal{R}_{\tau} = -\frac{1}{\tau} \mathbb{E}_{p_{\theta}} [(c^{\top} w - \bar{C}) w].$$

Difference between DDO-SPO+ and DDO-MD

Variant	What it preserves	What it borrows from RIPLM
DDO-SPO+ DDO-MD	SPO+ support-function surrogate soft decision distribution	smooth/compressed support moment centered residual mirror update

Presentation choice

For presenting to Adam, the safer main line is:

DDO-SPO+ first; DDO-MD as a related algorithmic variant.

What is established by construction

For DDO-SPO+:

1. The affine SPO+ terms stay unchanged.
2. Only the support function σ_C is regularized.
3. $\tau \downarrow 0$ recovers SPO+.
4. Strong convexity of Ω gives a smooth support selector.
5. Entropic smoothing gives an explicit soft-moment formula.

Point

These are clean mathematical facts, not empirical selling points.

What still needs proof

To make DDO a paper-level contribution, prove:

1. **Recovery:**

$$\ell_{\text{DDO}}^\tau \rightarrow \ell_{\text{SPO}+}.$$

2. **Bias control:**

$$\ell_{\text{DDO}}^\tau - \ell_{\text{SPO}+} = O(\tau).$$

3. **Gradient stability:** smoothness/Lipschitz bounds for $\nabla \sigma_{C,\tau}^\Omega$.

4. **Compression error:** moment approximation does not destroy the surrogate.

Compression-error target

Let the exact smoothed support be

$$\phi_{\tau,S}(u)$$

and the compressed approximation be

$$\phi_{\tau,Q}(u).$$

A useful theorem would bound

$$|\phi_{\tau,Q}(u) - \phi_{\tau,S}(u)| \leq \epsilon_{\tau}(u).$$

Then the DDO loss has only the additional error

$$\epsilon_{\tau}(c - 2\hat{c}).$$

Experimental mechanism to test

The experiment should test the mechanism, not just final regret.

Measure:

- test SPO regret;
- training surrogate value;
- gradient variance across minibatches;
- stability across seeds;
- number of solver calls;
- wall-clock time;
- temperature sensitivity;
- exact vs compressed moment gap.

Minimal first experiment

Use the classical shortest-path benchmark.

Start with small graphs where all paths can be enumerated.

Compare

SPO+ hard support selector vs. exact entropic DDO-SPO+.

Then move to larger graphs using soft dynamic programming or candidate moments.

Purpose

This separates smoothing from compression.

Ablations

Compare:

1. SPO+;
2. exact DDO-SPO+;
3. compressed DDO-SPO+;
4. candidate-set DDO;
5. DDO-MD residual variant;
6. fixed temperature vs annealing;
7. with and without AdaGrad-style residual scaling.

Goal

Identify which component matters.

What not to claim

Do not claim

DDO universally beats SPO+.

Do not claim

$O(N)$ for every downstream problem.

Correct claim

DDO is a regularized-support extension of SPO+ whose practical value depends on whether a good compressed support-moment oracle exists.

Clean science statement

Main statement

SPO+ is an affine term plus the support function

$$\sigma_C = \delta_C^*$$

evaluated at the dual direction $c - 2\hat{c}$.

DDO replaces this nonsmooth support function by the regularized conjugate

$$(\delta_C + \tau\Omega)^*.$$

This yields a smooth mirror moment map and recovers SPO+ as $\tau \downarrow 0$.

One-sentence version

Deep Decision Optimization is a smoothed and compressed Fenchel-dual extension of SPO^+ .

- It keeps the SPO^+ regret-surrogate backbone.
- It replaces the hard exposed-face subgradient with a regularized support moment.
- It borrows from RIPLM the idea that this moment can sometimes be computed in low-dimensional mirror coordinates.

References

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- Tiantian Zhang, *Rank-Induced PL Mirror Descent: A Rank-Faithful Second-Order Algorithm for Sleeping Experts*, arXiv:2509.18138.
- User-provided SPO-loss PDF: predict-then-optimize setup, geometric illustration, SPO+ surrogate, and Fisher-consistency motivation.